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COI Name (Institution):	Volpert, V. A. (Universite Lyon I) Wilke, H. (Institute of Crystal Growth)		
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Task Description:

Almost one hundred years ago Korteweg published a theory of how stresses could be induced in miscible fluids by concentration gradients, causing phenomena that would appear to be the same as with immiscible fluids. Miscible fluids could manifest a "transient" or "effective" interfacial tension (EIT). To this day, there has been no definitive experiment to confirm Korteweg's model but numerous fascinating and suggestive experiments have been reported.
The goal of this investigation is to answer the question: Can concentration and temperature gradients in miscible fluids induce stresses that cause convection? It is impossible to answer the question under 1 g conditions because the same gradients that cause the stresses also cause buoyant forces that overwhelm the Korteweg stresses.

In 1954 Cahn and Hilliard published their landmark thermodynamic analysis of diffuse interfaces in immiscible materials based on square gradient theory. Although they used different approaches, their mathematical results are similar to Korteweg's. Both approaches have a single parameter, k, and involve the square of the concentration gradients. We use the Cahn-Hilliard theory to extract the value of k for a monomer and its miscible polymer from spinning drop tensiometer experiments. Using this parameter in the Korteweg stress model with the Navier-Stokes equations, we predict transient, measurable fluid flows caused by externally-imposed gradients along a miscible interface between a monomer and its polymer. This experiment will constitute a definitive test of Korteweg's theory.

An interface between two miscible fluids can best be created via a spatially-selective photopolymerization, which allows the creation of precise and accurate concentration gradients between polymer and monomer. Optical techniques will be used to measure the refractive index variation caused by the resultant temperature and concentration fields. The convection induced by variations in the interfacial width can be studied independently from that caused by temperature and concentration gradients with a uniform interface width. Thus, the applicability of the square gradient energy term with a constant gradient energy parameter far from equilibrium can be tested and the relationship between the EIT measured by spinning drop tensiometry and Korteweg stresses confirmed.

Because the large concentration and temperature gradients cause buoyancy-driven convection that prevents the observation of the predicted flows, the experiment must be done in microgravity.

The results will not only put to rest a one hundred-year-old scientific mystery, they will provide important information on the application of the Cahn-Hilliard theory to miscible fluids. Confirmation of the numerical results will be of interest to all who use phase-field models or Cahn-Hilliard fluid models in simulating immiscible fluids. Knowledge gained from these experiments will permit evaluation of the role that concentration-gradient and temperature-gradient induced convection can play in industrial processing of miscible polymer systems. Finally, demonstrating that convection can occur in microgravity in miscible systems will be important to space processing of materials.

We supported a zero-upmass investigation, Miscible Fluids in Microgravity (MFMG) in which honey and water were the miscible fluids. The goal was to determine if such fluids could exhibit the Rayleigh-Tomotika instability and if a blob of water would become spherical. Also, the effect of a temperature gradient was studied to determine if drop migration would occur.

Because our flight investigation has been cancelled we have focused our efforts on spinning drop tensiometer to study the effective interfacial tension between isobutyric acid and water. This system has an Upper Critical Solution Temperature (UCST) of 26 °C. We study systems that are initially immiscible, i.e., below 26 °C, and then jump the temperature above the UCST and observe the behavior.

As part of the transition to the new exploration initiative, we have resurrected our relevant initial objectives from our investigation Frontal Polymerization in Microgravity, which was the original funded investigation that included questions about effective interfacial tension. Frontal polymerization is a localized polymerization reaction that propagates through the coupling of thermal transport and the strong rate dependence on temperature for an exothermic reaction.

The task is to demonstrate that frontal polymerization can be used for in space repair and fabrication and for in situ resource utilization. To these ends we are investigating how low of a temperature can frontal polymerization exist, considering the extremes of temperature that exist in low earth orbit and on the moon. We are also developing formulations based on commercial fillers, like clays, with multifunctional acrylates that could be readily applied to a hole under weightlessness and then rapidly cured as a front.

The final task is to develop monomers with zero vapor pressure for use in a human habitation. Current monomers vaporize to some extent. In some cases this creates an unpleasant odor and in other cases it could present an unacceptable level of contamination.

Determine if concentration and temperature gradients in miscible fluids induce stresses that can cause convection.

Develop frontal polymerization systems for in-space repair and manufacturing and for in-situ resource utilization.

There have been numerous reports of effects in miscible fluids that have been explained in terms of a transient, or effective, interfacial tension caused by sharp but transient concentration gradients. However, to our knowledge, no one has studied if convection could be caused by gradients in this EIT. Our work predicts that convection analogous to surface-tension induced convection (STIC) can occur at the transient interface of two miscible fluids, a phenomenon we refer to as "Effective Interfacial Tension Induced Convection (EITIC)." Verification of our predictions for fluid flow will be an exciting and novel finding.

Secondly, demonstrating the existence of this phenomenon in miscible fluids will open up a new area of study for materials science. Many polymer processes involving miscible monomer and polymer systems could be affected by fluid flow and so this work could help understand miscible polymer processing, not only in microgravity, but also on Earth.

The Cahn-Hilliard theory of diffuse interfaces is also based on the square gradient term of the free energy and yet the significance to miscible fluids is not clear. Our experiment directly tests the applicability of the square gradient energy to miscible systems. Not only does it relate to polymer processing but also the understanding of all phenomena analyzed in terms of a square gradient theory.

The modeling techniques we use can be applicable to a wide array of problems with and without interfaces.

The frontal polymerization research for space-based use can be directly applied to industrially-relevant problems.

	Monomers without vapor pressure could be useful on earth for applications in which organic vapors cannot be tolerated.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	We demonstrated conclusively that an effective interfacial tension can exist between miscible fluids. Isobutyric acid and water have an Upper Critical Solution Temperature (UCST) of 26.3 °C. We create a drop of IBA in water below the UCST and then increase the temperature above it. Long after the fluids have reached thermal equilibrium, the drop persists. The drop contracts when the rotation rate is decreased, indicating an interfacial tension exists. When the rotation rate is dropped from 7000 rpm to 1500 rpm, the drop breaks up via the Rayleigh-Tomotika instability. We calculate the interfacial tension from the Vonnegut formula. We have formulated multifunctional acrylate systems with peroxide and various fillers that can fill holes and propagate at low temperatures. We developed the first frontal thiol-ene system and demonstrated that adding a small amount of a trithiol to a diacrylate system can increase the front velocity 80%. We developed the first frontal polymerization system in which the peroxide initiator was encapsulated, which has the potential to increase the shelf life of the formulations. Four runs of the zero-upmass investigation, Miscible Fluids in Microgravity (MFMG), were performed on the ISS. The goal of MFMG is to determine if interfacial phenomena seen with immiscible fluids could be seen with miscible fluids. The experiments had to be performed with existing materials on the ISS. Honey and water were chosen as the fluids, and urine collection syringes were used as the vessels in which the experiments were performed. In March (Increment 8) Dr. Michael Foale performed four experiments under isothermal conditions to determine: If a stream of honey injected into water would exhibit the Rayleigh-Tomotika instability and break into small drops. If an aspherical drop of water in honey would spontaneously assume a spherical shape. The experiments were performed successfully. No behavior beyond simple diffusion was observed, which is allowing us to estimate the maximum possible value of the square gradient parameter in our model with the Navier-Stokes equations plus a Korteweg stress term. During Increment 9, Mike Fiske performed two runs in which a stream of honey was injected into water while the syringe was attached to the surface of the Commercial Generic Bioprocessing Apparatus (CGBA) at approximately 31 °C. Preliminary analysis indicates that some fluid motion occurred. It is possible that the apparent migration of the stream was not caused by residual buoyancy-induced convection and therefore may be an indication that Korteweg stresses can be important in miscible fluids.
Bibliography Type:	Description: (Last Updated: 09/26/2012)
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